PICU Up!: Impact of a Quality Improvement Intervention to Promote Early Mobilization in Critically Ill Children*

Beth Wieczorek, DNP, PNP-AC[1]; Judith Ascenzi, DNP, RN, CCRN[2]; Yun Kim, MS, OTR/L[3]; Hallie Lenker, PT, DPT, STAR/C[3]; Caroline Potter, MS, CCLS, CIMI[4]; Nehal J. Shata, MBBS[1]; Lauren Mitchell, MS, CCLS[4]; Catherine Haut, DNP, CPNP-AC,CCRN[5,6]; Ivor Berkowitz, MBBCh, MBA[1,7]; Frank Pidcock, MD[3,7]; Jeannine Hoch, MA, CCC-SLP[7]; Connie Malamed, MA[8]; Tamara Kravitz, MS[8]; Sapna R. Kudchadkar, MD[1,7]

Objective: To determine the safety and feasibility of an early mobilization program in a PICU.

Design: Observational, pre-post design.

Setting: PICU in a tertiary academic hospital in the United States.

Patients: Critically ill pediatric patients admitted to the PICU.

Intervention: This quality improvement project involved a usual-care baseline phase, followed by a quality improvement phase that implemented a multicomponent, interdisciplinary, and tiered activity plan to promote early mobilization of critically ill children.

Measurements and Main Results: Data were collected and analyzed from July to August 2014 (preimplementation phase) and July to August 2015 (postimplementation). The study sample included 200 children 1 day through 17 years old who were admitted to the PICU and had a length of stay of at least 3 days. PICU Up! implementation led to an increase in occupational therapy consultations (44% vs 59%; $p = 0.034$) and physical therapy consultations (54% vs 66%; $p = 0.08$) by PICU day 3. The median number of mobilizations per patient by PICU day 3 increased from 3 to 6 ($p < 0.001$). More children engaged in mobilization activities after the PICU Up! intervention by PICU day 3, including active bed positioning ($p < 0.001$), and ambulation ($p = 0.04$). No adverse events occurred as a result of early mobilization activities. The most commonly reported barriers to early mobilization after PICU Up! implementation was availability of appropriate equipment. The program was positively received by PICU staff.

Conclusions: Implementation of a structured and stratified early mobilization program in the PICU was feasible and resulted in no adverse events. PICU Up! increased physical therapy and occupational therapy involvement in the children's care and increased early mobilization activities, including ambulation. A bundled intervention to create a healing environment in the PICU with structured activity may have benefits for short- and long-term outcomes of critically ill children. (*Pediatr Crit Care Med* 2016; 17:e559–e566)

Key Words: acute rehabilitation; children; early mobilization; intensive care units; sleep

***See also p. 1194.**

[1]Department of Anesthesiology and Critical Care Medicine, Charlotte R. Bloomberg Children's Center, Johns Hopkins University School of Medicine, Baltimore, MD.

[2]Department of Pediatric Nursing, Charlotte R. Bloomberg Children's Center, Johns Hopkins Hospital, Baltimore, MD.

[3]Department of Physical Medicine and Rehabilitation, Johns Hopkins University School of Medicine, Baltimore, MD.

[4]Department of Child Life, Charlotte R. Bloomberg Children's Center, Johns Hopkins Hospital, Baltimore, MD.

[5]University of Maryland School of Nursing, Baltimore, MD.

[6]Pediatrix Medical Group, Herman-Walter Samuelson Children's Hospital at Sinai, Baltimore, MD.

[7]Department of Pediatrics, Charlotte R. Bloomberg Children's Center, Johns Hopkins Hospital, Baltimore, MD.

[8]Interactive Learning, Johns Hopkins Medicine, Baltimore, MD.

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For information regarding this article, E-mail: skudcha1@jhmi.edu

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The central focus of care in the PICU is on resuscitation, stabilization, management of critical disease processes, and reversal of organ failure. As a result, children are often sedated, restrained, and confined to bed for prolonged periods due to perceived benefits of safety, comfort, and hemodynamic stability (1). A culture of immobility may have several short- and long-term implications for critically ill children and their families (2–4), negatively affecting circadian rhythms and potentially increasing the risk of delirium (5–10). Ineffective positioning and limited mobility contribute to a high frequency of pressure ulcers in both infants and children (11–13). Furthermore, children who survive their illness can experience residual perceptual-motor, psychiatric, and behavior problems as sequelae of critical illness (8).

Data from adult ICUs demonstrate that structured and interdisciplinary approaches to early mobilization are associated with reduced intensive care and hospital length of stay, improved muscle strength and self-perception of functional status, and decreased sedation, delirium, and length of mechanical ventilation (14–17). Although pediatric literature regarding early mobilization practices is just emerging, available data indicate that early mobilization activities for the critically ill child are likely safe and feasible and may have short- and long-term benefits (18–21).

The overall objective of this quality improvement (QI) project was to evaluate a structured and interdisciplinary early rehabilitation and progressive mobility protocol, "PICU Up!," for all critically ill children in a large, academic, tertiary-care PICU. Previously, the referral of children for therapy and mobilization was at the discretion of the medical providers without standardization or guidelines. The goals of PICU Up! were to provide a standardized, evidence-based, interdisciplinary mechanism to increase each child's activity level safely while simultaneously initiating and promoting a culture of mobility.

MATERIALS AND METHODS

Overview of Project Setting, Design, and Patients

The PICU Up! initiative was developed using an established and structured QI framework (described below in "QI Process") and evaluated using a pre-post design (22). The setting was the Johns Hopkins PICU, a 40-bed academic, tertiary care, combined medical-surgical unit with single-patient rooms that provides care for children from 1 day to 21 years old. Registered nurse-to-patient ratios are 1:1 or 1:2 depending on patient acuity. Pediatric occupational therapist (OT), physical therapist (PT), and speech/language pathologist consultations and treatments are available when ordered by a medical provider. The PT and OT staff who care for children in the PICU also staff other units in the hospital. PICU Up! was implemented without any additional personnel or equipment resources.

A nonprobability, convenience sampling strategy was used for program evaluation. Inclusion criteria were children 1 day to 17 years old who required PICU admission for three or more days. Exclusion criteria included extracorporeal membrane oxygenation (ECMO), open chest or abdomen, unstable fractures, or medical orders specifying alternate activities. Charts of the first consecutive 100 children who met inclusion criteria were reviewed retrospectively both before (July to August 2014) and after (July to August 2015) implementation of the mobility protocol. PICU Up! was implemented in March to May 2015. This QI project was comprehensively reviewed and acknowledged by the Johns Hopkins Institutional Review Board (IRB) as "QI," and did not require IRB approval.

QI Process

The structured QI model used for the PICU Up! initiative aimed to create a change in practice through the "four Es" approach: engage, educate, execute, and evaluate (22). To achieve this, a collaborative interdisciplinary QI team, the PICU Up! Working Group, was created. Participation in the PICU Up! Working Group was open to all interested PICU staff. Active participants included at least one champion from each of the following groups: physicians, nurse practitioners, nursing, OT, PT, child life specialists, speech language pathology, respiratory therapy, and rehabilitation medicine. This core group of individuals met together on a weekly basis over 18 months to plan (engage and educate) the QI project prior to its execution and evaluation.

First, champions from the interdisciplinary team conducted focus groups with all stakeholders from their respective disciplines to present the problem and identify PICU-based facilitators and barriers to early mobilization. Second, the group worked together to develop guidelines for mobilization in the PICU based on the feedback provided from the focus groups and prior literature regarding early mobilization (14, 22). Third, educational resources were developed for all PICU staff (described below in "Staff Educational Resources"). Finally, the interdisciplinary team determined valid and feasible outcome measures based on previously published adult early mobilization QI studies to evaluate performance.

The PICU Up! Program

Components of PICU Up! included a tiered activity plan based on clinical variables and exclusion criteria (**Table 1**). A foundation of PICU Up! was implementation of sleep hygiene promotion and routine delirium screening for all children. Each activity level was connected with a set of interventions, written to promote individualization based on the child's unique needs. Also included were criteria for pausing activity and reassessing the child before continuing the activity (**Table 2**).

It was determined that each child's activity level would be discussed and established during morning rounds, with the bedside nurse reporting the PICU Up! level (1, 2, or 3) based on the established criteria (Table 1). The interdisciplinary team caring for the patient considered the activity level and activity goal for the day during rounds. The bedside nurse recorded this information in the electronic record daily goals note, which was accessible to all providers, and assessed the appropriateness of the activity level throughout the day. Activities could be implemented by nursing and/or therapists based on their specific skills and the child's needs. The PICU Up! levels were intended to provide guidelines for appropriate activities, with the option to perform more advanced activities (i.e.,

TABLE 1. PICU Up! Levels and Tiered Activity Plan

Excluded from PICU Up! levels and activities	Extracorporeal membrane oxygenation Open chest Open abdomen Unstable fracture Medical orders specifying alternate activity	
PICU Up! Level	**Parameters for Inclusion**	**Activities**
Level 1	Intubated with $F_{IO_2} > 60\%$ or Intubated with PEEP > 8 or Intubated difficult airway or New tracheostomy or Acute neurologic event or Sedated and SBS −3 to −2 or Vasopressor other than milrinone	Lights on/shades up by 09:00 Bed/bath/weight by 23:00 Lights dimmed/out by 23:00 increase lighting as needed for cares/interventions Television limited to 30 min at a time and a goal of < 2 hr/d for children > 2 yr old Head of bed elevated ≥ 30° Turn every 2 hr during the day and every 4 hr at night Positioned in developmentally supportive position or as recommended by OT/PT OT consult by PICU day 3 PT consult as needed
Level 2	Intubated or tracheostomy with $F_{IO_2} \leq 60\%$ +/or PEEP ≤ 8 and SBS −1 to +3 or Noninvasive support with $F_{IO_2} > 60\%$ or Dialysis/renal replacement therapy or Femoral access	Level 1 activities plus Positive touch for infants/toddlers Sitting up in bed TID Team to consider OOB to chair +/or ambulation OT/PT consult by PICU day 3 Assess for difficulty with communication or phonation and consult SLP Assess for swallowing readiness in high risk children and consult SLP Assess need for daily schedule Preschool Confusion Assessment Method-ICU or Pediatric Confusion Assessment Method-ICU BID
Level 3	Noninvasive pulmonary support with $F_{IO_2} \leq 60\%$ or Baseline pulmonary support or External ventricular drain cleared by neurosurgery and SBS −1 to +3	Level 1 and 2 *plus* OOB to chair TID or sitting up in bed TID if appropriate chair is not available Ambulate BID if trunk control present

BID = twice a day, OOB = out of bed, OT = occupational therapist, PEEP = positive end-expiratory pressure, PT = physical therapist, SBS = State Behavioral Scale, SLP = speech language pathology, TID = three times a day.

ambulation of an intubated patient) with interdisciplinary team discussion individualized to the child.

Staff Educational Resources

Educational resources were presented in meetings, presentations, and online communications to ensure involvement of all PICU stakeholders. A formal written procedure was available to staff via the Johns Hopkins online policies portal. Formal educational materials included a required 10-minute online learning module about early rehabilitation in critically ill patients, including a brief overview of relevant adult and pediatric literature. The module then transitioned into interactive case-based scenarios to illustrate application of the PICU Up! criteria (**Supplemental Fig. 1**, Supplemental Digital Content 1, http://links.lww.com/PCC/A326—**legend**, Supplemental Digital Content 7, http://links.lww.com/PCC/A332; **Supplemental Fig. 2**, Supplemental Digital Content 2, http://links.lww.com/PCC/A327—legend, Supplemental Digital Content 7, http://links.lww.com/PCC/A332; **Supplemental Fig. 3**, Supplemental Digital Content 3, http://links.lww.com/PCC/A328—legend, Supplemental Digital Content 7; and **Supplemental Fig. 4,** Supplemental Digital Content 4, http://links.lww.com/PCC/

TABLE 2. Criteria to "Rest and Reassess" From PICU Up! Pocket Card Distributed to All Staff

Change in heart rate, 20%
Change in blood pressure, 20%
Change in respiratory rate, 20%
Decrease arterial oxygen saturation, 15%
Increase F_{IO_2}, 20%
Increase end-tidal CO_2, 20%
Ventilator asynchrony
Continuous/bilevel positive airway pressure asynchrony
Respiratory distress
New arrhythmia
Hemodynamic concerns
Change in mental status
Concern for airway device, vascular access, or external ventricular drain integrity
Behavior interfering with safe activity

Vital sign changes based on baseline value prior to activity initiation.

A329—legend, Supplemental Digital Content 7, http://links.lww.com/PCC/A332), which demonstrate components of the online module. Because we identified a simultaneous need for pediatric delirium education among the staff (23), a parallel educational module for pediatric delirium utilizing the same interactive cases was developed. A pocket card that summarized the levels and activities was distributed to all staff members.

PICU Outcome Measures

Early mobilization was defined by the team as any passive or active activity that occurred within the first 3 days of PICU admission and was intended to maintain or restore musculoskeletal strength and function. At our institution, both PT and OT must be formally consulted by physicians or nurse practitioners to work with patients. PTs facilitate ambulation and mobility, while splinting, range of motion, seating, and positioning are the purview of OTs, with PT collaboration as needed. The primary outcome measures were as follows: 1) the proportion of patients with OT and/or PT consultations by PICU day 3 and 2) the number and types of mobilization activities performed by PICU day 3. Activities were categorized as in-bed therapies or mobility therapies. In-bed therapies included passive range of motion, active range of motion, active or passive bed positioning, and splinting. Mobility activities included sitting at edge of bed, sit to stand, transfer, ambulation, and play. Secondary outcome measures were as follows: 1) the number of times and reasons that activities were stopped; 2) barriers to activities; and 3) mobilization-related adverse events, including inadvertent extubation or line removal. Data for all outcome measures, including mobilization activities, adverse events, and barriers were collected for each day of PICU admission in all patients. After implementation, a Likert-style four-question, voluntary and anonymous response questionnaire was sent to the entire PICU staff to collect general feedback about PICU Up! and ongoing barriers to mobilizing the critically ill child.

Statistical Analysis

Descriptive statistics were summarized as proportions for binary and categoric data. Continuous variables are presented as means ± SD or as medians with interquartile ranges (IQRs), unless otherwise indicated. Unadjusted comparisons before and after implementation of PICU Up! were performed by using chi-square analysis, Fisher exact test, or the Wilcoxon signed rank test, as appropriate. A two-sided p value less than 0.05 was used to determine statistical significance. All data were analyzed with the statistical software package STATA version 12.0 (StataCorp LP, College Station, TX).

RESULTS

Patient Characteristics

Data were collected from 100 children who met inclusion criteria in the preimplementation group (675 PICU days) and 100 children in the postimplementation group (737 PICU days). Characteristics of the pre- and postimplementation groups are summarized in **Table 3**. The groups were similar with regard to age, weight, PICU admission diagnoses, and Pediatric Risk of Mortality scores. The PICU Up! level did not differ between pre- and postimplementation groups at PICU day 3 ($p = 0.79$).

Occupational and Physical Therapy Consultations

OT consultations increased significantly from the pre- to postimplementation phase by PICU day 3 (44% vs 59%; $p = 0.034$). The number of PT consultations by PICU day 3 increased but did not reach significance (54% vs 66%; $p = 0.08$).

Activities

The total number of PICU mobilization activities in the first 3 days of admission was 465 preimplementation and 769 postimplementation, respectively. As shown in **Table 4**, by PICU day 3, the median number of mobilization activities per patient before and after PICU Up! implementation doubled from 3 to 6 (IQR, 2–5 and 3–7.5, respectively; $p < 0.001$). The proportion of children receiving at least one in-bed activity increased significantly from 70% to 98% ($p < 0.001$). Although there was no change in passive range of motion, passive bed positioning, splinting, or active range of motion, there was a significant increase in the number of children receiving active bed positioning (26% vs 57%; $p < 0.001$).

More children engaged in at least one mobility activity by day 3 after PICU Up! implementation (63–76%; $p = 0.05$). However, there was no increase in the proportion of children who participated in sitting on the edge of bed, sit to stand, transfer, or play. After PICU Up! implementation, 27% of all children had ambulated by day 3, increased from 15% preimplementation ($p = 0.04$). Among children greater than or

TABLE 3. Patient Characteristics Before and After PICU Up! Implementation

Characteristic	Preimplementation (n = 100)	Postimplementation (n = 100)	p[a]
Age, yr, mean (SD)	7.7 (5.6)	7.7 (5.4)	0.99
Weight, kg, mean (SD)	26.8 (19)	27.3 (22)	0.86
Gender, *n*			0.02[b]
Male	67	51	
Female	33	49	
Admission categories, *n*			0.26
Medical	58	50	
Surgical	42	50	
Intubated on admission	39	46	0.32
Preexisting conditions, *n*			
Motor impairment	29	26	0.63
Intellectual disability	32	27	0.44
Pediatric Risk of Mortality score, mean (SD)	4.9 (4.4)	5.4 (4.5)	0.36
PICU length of stay, mean (SD)	6.8 (5.4)	7.6 (6.9)	0.34
PICU Up! level—day 3, *n*			0.79
1	7	11	
2	18	17	
3	64	60	
Excluded	11	12	

[a]Descriptive, *t* test, and chi-square tests used for analysis.
[b]Statistically significant.

equal to 3 years old, 20% (10/50) ambulated prior to PICU Up! implementation, whereas 39% (22/56) ambulated after implementation ($p = 0.03$). None of the 39 orally intubated children ambulated prior to PICU Up!, whereas four of 40 orally intubated children (10%) ambulated after PICU Up! implementation (**Supplemental Fig. 5**, Supplemental Digital Content 5, http://links.lww.com/PCC/A330—legend, Supplemental Digital Content 7, http://links.lww.com/PCC/A332).

Barriers to Activities and Adverse Events

Barriers to activities before and after PICU Up! implementation in the first 3 days of PICU admission are shown in **Table 5**. Preimplementation, the most frequently reported barrier to early mobilization activities as documented by nursing and/or therapists included procedures (n = 19), patient condition (n = 10), and bed rest orders (n = 3). Postimplementation, the most frequently reported barrier was lack of equipment, specifically age and size-appropriate seating devices and positioning materials (n = 22), which was reported only twice preimplementation. Procedures were less frequently reported as a barrier (n = 10) after PICU Up! implementation and patient condition continued to be reported with similar frequency (n = 11).

No adverse events related to mobility, including unplanned extubations or line displacements, were reported in either phase of the program. In addition, there were no documented events of aborting mobilization activities once they had been initiated for hemodynamic, respiratory, pain, behavior, or pain criteria.

Feedback From PICU Staff

Ninety-five PICU staff responded to the postimplementation feedback questionnaire, and 58% of respondents were nurses. Most staff (95%) indicated that the PICU Up! levels were helpful in planning activities. Seventy-three percent reported that the pocket card was the most useful resource for daily use to increase knowledge regarding mobilization of the critically ill child, and 68% agreed that they had the support needed from team members to increase patient activity. Support from PTs and OTs was reported as a valuable personnel resource for mobilization. Barriers included not having adequate staff to mobilize the child while continuing to care for the other patients. Respondents also indicated that the medical team was not always sensitive to competing demands of the bedside caregivers when discussing mobilization.

TABLE 4. Early Mobilization Activities: First 3 Days of PICU Admission

Activity (No. of Children Participating in That Activity)	Preimplementation $n = 100$	Postimplementation $n = 100$	p[a]
In-bed activities, *n*			
Passive range of motion	13	17	0.43
Passive bed positioning	41	47	0.39
Splinting	3	9	0.08
Active range of motion	2 (2)	2 (1)	0.99
Active bed positioning	26	57	< 0.001[b]
At least one bed activity	70	98	< 0.001[b]
Mobility activities, *n*			
Sit edge of bed	6	11	0.20
Sit to stand	24	30	0.34
Transfer	48	46	0.77
Ambulate	15	27	0.04[b]
Play	6	3	0.78
Other	5	3	0.31
At least one mobility activity	63	76	0.05[b]

[a]Descriptive, Fisher exact, and chi-square tests used for analysis, as appropriate.
[b]Statistically significant.

TABLE 5. Barriers to Activities: First 3 Days of PICU Admission

No. of Times Barrier Reported	Preimplementation	Postimplementation	p[a]
Barrier			< 0.001
Child refused	0	3	
Parent refused	1	3	
Test/study/procedure/surgery	19	10	
Patient condition	10	11	
Equipment availability	2	22	
Bed rest order	3	0	

[a]Fisher exact test used for analysis.

DISCUSSION

Through a structured QI process, the PICU Up! Early Rehabilitation and Progressive Mobility Program established an interdisciplinary, multicomponent, and stratified intervention for critically ill children admitted to the PICU. Implementation of a bundled intervention to create a culture of mobility in the PICU was safe and feasible and facilitated the formal involvement of PTs and OTs in the care of critically ill children. Although PICU Up! led to increases in early mobilization activities of critically ill children, our QI initiative identified several areas for ongoing education and modification.

A structured and stratified approach to adult ICU rehabilitation has been shown to increase the success of mobility program implementation (14, 16, 24–28). In line with the adult literature, key factors that fostered a culture of mobility after PICU Up! implementation included the following: 1) standardization of workflow, 2) facilitation of discussions regarding mobility goals during morning work rounds, and 3) identification of safe activities for each patient based on his/her acuity level and specific needs. Unlike the adult literature, data to support early rehabilitation in the PICU are limited to pilot studies, case series, and case reports (21). To our knowledge, this is the first report of a large-scale, multifaceted QI intervention aimed at promoting a culture of mobility in the PICU setting.

PICU Up! significantly increased the formal involvement of OTs and PTs in the care of critically ill children. These resources were determined to be the most valuable by all staff in creating a culture of PICU mobility. Like many adult and PICUs nationally, our PICU does not have dedicated PT/OT staff (29), and because PTs and OTs care for children throughout the hospital, they must be formally consulted by physicians or nurse practitioners to work with PICU patients. Therefore, prior to implementation, we observed that PT/OT consultations were commonly delayed for the most critically ill children. This observation is consistent with a recent Canadian survey in which physiotherapists identified the need for a physician order as the major barrier to PICU rehabilitation (30). Before PICU Up! implementation, even when PT/OT consults were ordered early, nurses and/or physicians often would turn away therapy staff because they perceived that the patient's acuity was a contraindication to rehabilitation therapies. This was true particularly for intubated children. In addition to creating a framework for timely PT/OT consultation for all PICU patients, educating staff about the breadth of early mobilization activities and empowering PT/OT staff to advocate for therapy was crucial. Nurses also used PT/OT sessions to learn simple interventions from therapists that they could facilitate between sessions. Guidelines for stopping and reassessing children during therapy activities also enabled all staff involved to have a shared mental model.

The increase in OT consultations by PICU day 3 is consistent with prior pre-post implementation mobility studies investigating early activity in ICUs. Unlike programs demonstrating increased mobility activities as measured by billable PT units, we did not observe a significant increase in PT referrals by PICU day 3 (31). However, more children ambulated, suggesting that PICU Up! facilitated a transition to early ambulation through streamlined workflow and discussions.

Like other pediatric studies, our large-scale QI intervention did not result in any mobilization-related adverse events, such as inadvertent extubation (18–20). There were also no reports of a child needing to terminate activities early due to a decline in physiologic status or behavior. This finding is likely related to the fact that we designed the mobility levels based on the physiologic and developmental level of each child.

Barriers to implementation of early mobilization programs have been well-documented in the adult literature (17). For this QI study, additional resources were not allotted for personnel or equipment, and consistent with the adult ICU experience, PICU staff indicated through documentation and feedback that lack of appropriate equipment was a major deterrent to mobilizing patients after PICU Up! implementation. Although the PICU Up! program provided a streamlined approach to early mobilization, specific activities could not be performed despite multidisciplinary team planning on morning rounds due to equipment barriers. Equipment challenges are further magnified in the pediatric population because of the heterogeneous nature of patients' heights and weights. Procedures were reported less frequently as a barrier after PICU Up!, suggesting that team coordination may improve the timing of early mobilization activities.

Finally, the exclusion criteria for PICU Up! were based on several factors. First, in order to ease the implementation phase, address safety concerns, and optimize acceptability of the program among staff, we chose to exclude ECMO patients in the initial PICU Up! rollout given the unique needs of this patient population. After the completion of this study, we have begun translating the PICU Up! principles to ECMO patients and are successfully increasing mobilization activities in this population. (**Supplemental Fig. 6**, Supplemental Digital Content 6, http://links.lww.com/PCC/A331—legend, Supplemental Digital Content 7, http://links.lww.com/PCC/A332). The PICU Up! criteria will be modified after a pilot phase with ECMO-specific activity guidelines developed in conjunction with the ECMO team. Second, postsurgical patients with specific orders outlining mobilization restrictions were excluded from the initial implementation, although sleep hygiene and delirium screening were in place for all patients. The same applied for open chests and open abdomens. All in all, the vast majority of surgical patients were included in the PICU Up! program which was well-received by our surgeons.

This study had several important limitations. First, data collected from a convenience sample in a single academic pediatric hospital may limit the generalizability. Second, the retrospective nature of data collection is limited by the quality of documentation, specifically with regards to barriers. Third, only children who remained in the PICU for three or more days were included, limiting generalizability for short-stay patients. Fourth, we were unable to determine changes in delirium incidence given that delirium screening was not yet routine in the preimplementation phase. Fifth, we did not measure the impact of PICU Up! implementation on resource utilization and workload, an important future direction for research given that most PICUs do not have dedicated PTs and OTs. Finally, data on sedative-analgesic dosing were not collected.

CONCLUSION

A multicomponent and interdisciplinary approach to early mobilization of children admitted to the PICU is feasible and resulted in no adverse events. PICU Up! increased the formal involvement of PTs and OTs and increased early mobilization activities including ambulation. Promoting and maintaining a culture of mobility in the PICU creates a strong foundation to investigate the effects of early mobilization on functional outcomes in critically ill children to potentially improve short- and long-term outcomes while providing a healing PICU environment.

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